

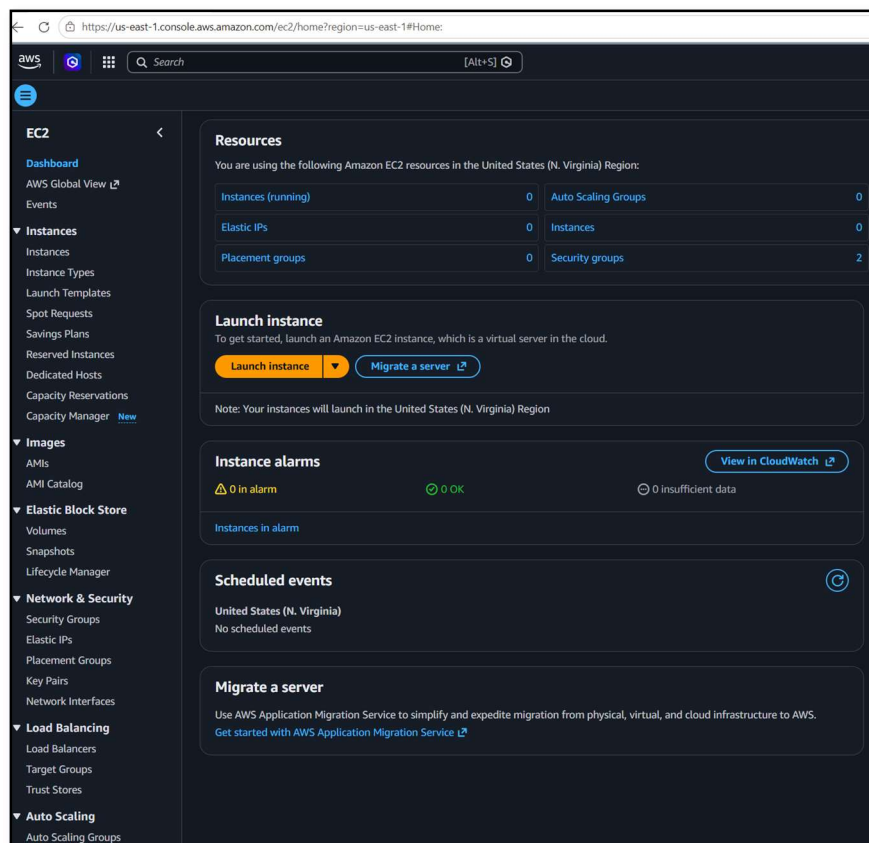
## Lab 2: Building a Resilient & Automated Web Service (5 Marks)

**Objective:** Deploy a fault-tolerant, auto-scaling web application using an Application Load Balancer (ALB), an Auto Scaling Group (ASG), and an automated launch template.

### Part 1: Automating Web Server Setup with a Launch Template

1. **Navigate to Launch Templates:**
  - 1.1 Open the **EC2 Dashboard**

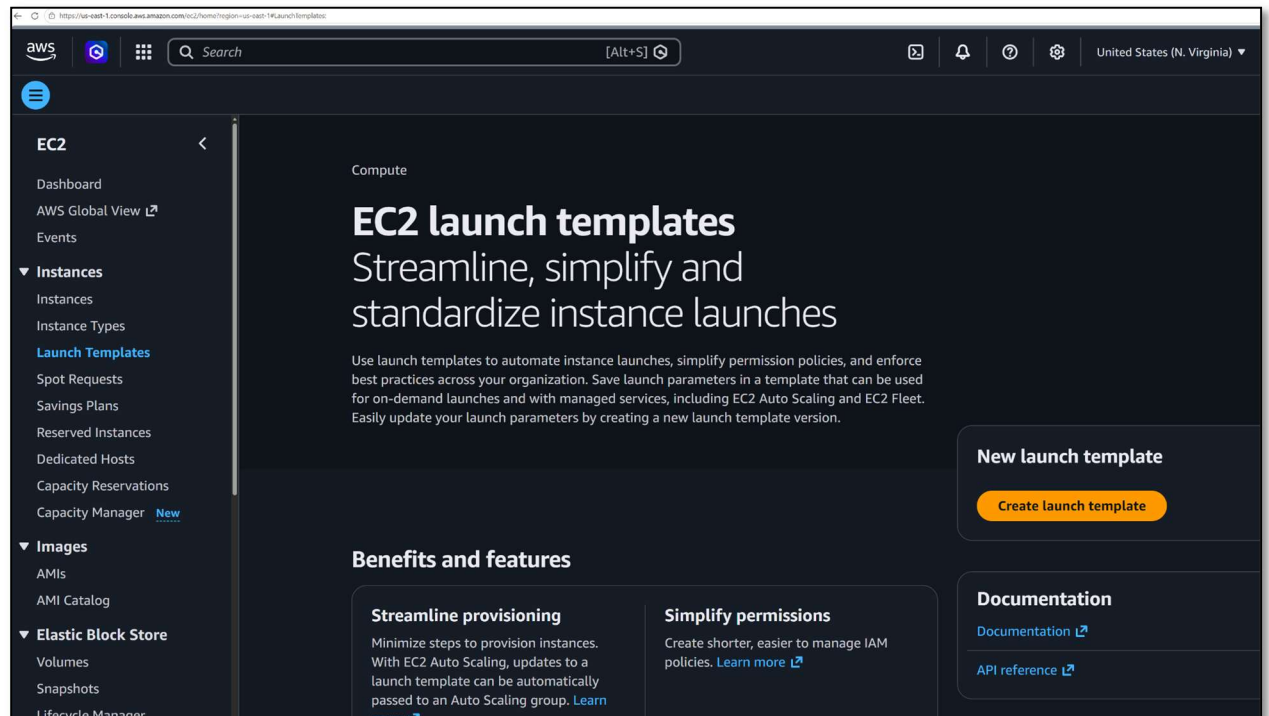
#### EC2 Dashboard



**Note:** Shows the entry point to EC2 resources before opening Launch Templates.

1.2 Select **Launch Templates** from the left-hand menu.

### Launch Templates Page



**Note:** Shows the Launch Templates service and the option used to create a new template.

1.3 Click **Create launch template**.

### Create Launch Template Page

The screenshot displays the 'Create launch template' form in the AWS Management Console. The breadcrumb trail at the top indicates the path: 'EC2 > Launch templates > Create launch template'. The form is titled 'Create launch template' and includes a brief description. It is divided into two main sections. The left section, 'Launch template name and description', contains two text input fields: 'Launch template name - required' (with the value 'MyTemplate') and 'Template version description' (with the value 'A prod webserver for MyApp'). Below these fields is a checkbox for 'Auto Scaling guidance' which is currently unchecked. The right section, 'Summary', lists the configuration details: 'Software Image (AMI)', 'Virtual server type (instance type)', 'Firewall (security group)', and 'Storage (volumes)'. At the bottom right, there are two buttons: 'Cancel' and 'Create launch template'.

**Note:** Shows the form used to define a reusable EC2 instance configuration.

2. **Template Metadata & Tagging (Student Verification Step):**
  - 2.1 Launch template name: WebServer-Template-[YourLastName]-[StudentID]
  - 2.2 Template version description: Initial automated web server configuration

### Launch Template Metadata

aws | [Search] [Alt+S]

EC2 > Launch templates > Create launch template

Launch template name and description

Launch template name - *required*

WebServer-Template-Hossain-567965

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '\*', '@'.

Template version description

Initial automated web server configuration

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

▶ Template tags

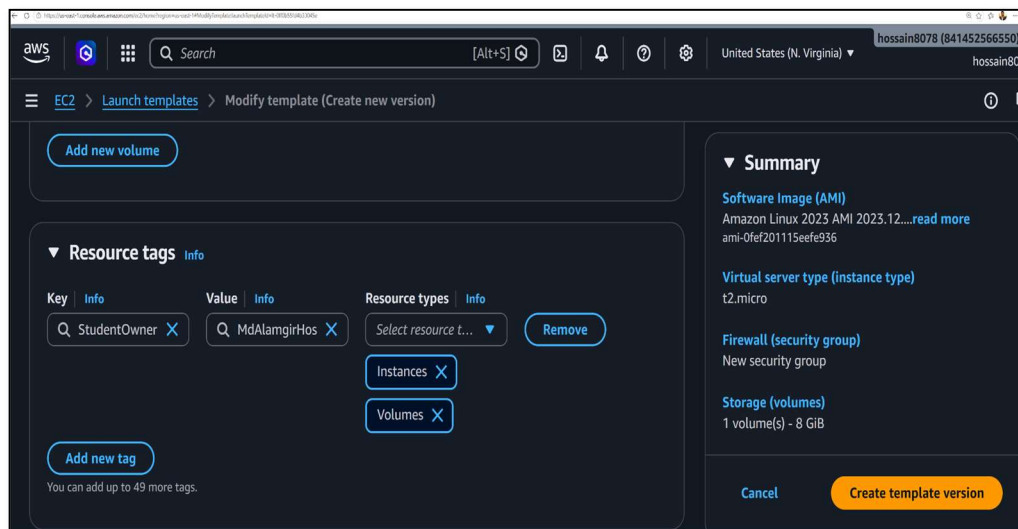
▶ Source template

**Note:** Confirms the personalized launch template name and version description.

2.3 Under **Resource tags**, click **Add tag**:

- **Key:** StudentOwner
- **Value:** [YourFullName]-[YourStudentID]
- **Resource types:** Select **Instances, Volumes**

### Resource Tags



**Note:** Confirms the StudentOwner tag is applied to instances and volumes.

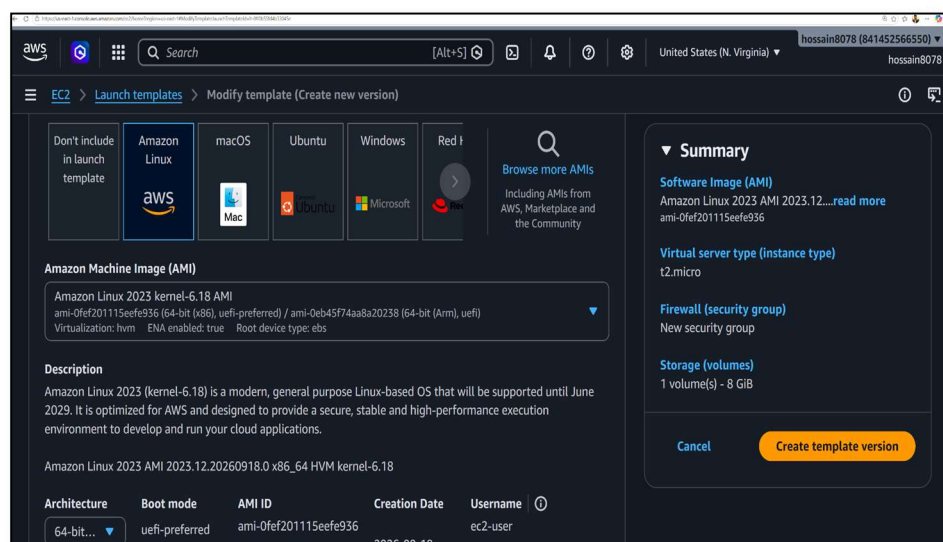
- **Launch templates.**

3. **Application and OS Images (AMI):**

3.1 Under **Quick Start**, select **Amazon Linux**.

3.2 Under the **AMI dropdown**, confirm **Amazon Linux 2023 AMI (HVM)** is selected (x86\_64 architecture, Free tier eligible).

### Amazon Linux 2023 AMI



**Note:** Confirms the Amazon Linux 2023 x86-64 image selected for the web servers.

#### 4. Instance Type & Key Pair:

**4.1 Instance Type:** Select t2.micro (or t3.micro, depending on regional Free Tier availability).

**4.2 Key pair (login):** Select the key pair you created in Lab 1.

#### Instance Type and Key Pair

The screenshot displays the AWS Management Console interface for configuring an EC2 instance template. The top navigation bar shows the AWS logo, a search bar, and the current region (United States (N. Virginia)). The breadcrumb trail indicates the path: EC2 > Launch templates > Modify template (Create new version). The main content area is divided into two columns. The left column contains the 'Instance type' section, which is expanded to show the 't2.micro' instance type. Below this, there is a list of pricing information for various operating systems. The right column contains the 'Summary' section, which lists the 'Software Image (AMI)' as 'Amazon Linux 2023 AMI 2023.12...', the 'Virtual server type (instance type)' as 't2.micro', and the 'Storage (volumes)' as '1 volume(s) - 8 GiB'. At the bottom of the console, there is a 'Key pair (login)' section with a text input field containing 'CCMP200T05' and a 'Create new key pair' button.

**Instance type** Info | Get advice Advanced

Instance type

t2.micro  
Family: t2 1 vCPU 1 GiB Memory Current generation: true  
On-Demand Linux base pricing: 0.0116 USD per Hour  
On-Demand Windows base pricing: 0.0162 USD per Hour  
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour  
On-Demand SUSE base pricing: 0.0116 USD per Hour  
On-Demand RHEL base pricing: 0.026 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

**Key pair (login)** Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

CCMP200T05 Template value Create new key pair

**Summary**

Software Image (AMI)  
Amazon Linux 2023 AMI 2023.12....read more  
ami-0fef201115ee9e936

Virtual server type (instance type)  
t2.micro

Firewall (security group)  
New security group

Storage (volumes)  
1 volume(s) - 8 GiB

Cancel Create template version

**Note:** Confirms the t2.micro instance type and the selected login key pair.

5. **Network Settings:**

5.1 Select **Create security group**.

5.2 **Security group name:** WebServer-SG-[YourLastName]

5.3 **Description:** Allow inbound HTTP and SSH traffic

5.6 **VPC:** Ensure your **Default VPC** is explicitly selected in the dropdown.

**New Security Group Configuration**

The screenshot displays the AWS Management Console interface for configuring a new security group. The main configuration area includes the following sections:

- Availability Zone:** A dropdown menu set to "Don't include in launch template" with a link to "Enable additional zones".
- Firewall (security groups):** A section with a description and two radio buttons: "Select existing security group" (unselected) and "Create security group" (selected).
- Security group name - required:** A text input field containing "WebServer-SG-Hossain".
- Description - required:** A text input field containing "Allow inbound HTTP and SSH traffic".
- VPC:** A dropdown menu showing "vpc-0ab664410caf1682b (default)".

On the right side, the **Summary** panel provides a overview of the configuration:

- Software Image (AMI):** Amazon Linux 2023 AMI 2023.12....read more
- Virtual server type (instance type):** t2.micro
- Firewall (security group):** New security group
- Storage (volumes):** 1 volume(s) - 8 GiB

At the bottom of the Summary panel, there are two buttons: "Cancel" and "Create template version".

**Note:** Confirms the personalized security group and default VPC configuration.

## 5.7 Inbound Rules:

- **Rule 1:** Type: **HTTP** | Port: 80 | Source: **Anywhere** (0.0.0.0/0).
- **Rule 2:** Type: **SSH** | Port: 22 | Source: **My IP**.

### Inbound Security Group Rules

The screenshot displays the AWS Management Console interface for configuring inbound security group rules. The top navigation bar shows the AWS logo, a search bar, and the current region (United States (N. Virginia)). The breadcrumb trail indicates the path: EC2 > Launch templates > Modify template (Create new version).

The main content area is titled "Inbound Security Group Rules". It contains two rules, each with a "Remove" button.

**Security group rule 1 (TCP, 80, 0.0.0.0/0)**

| Type | Protocol | Port range | Source type | Source    | Description - optional     |
|------|----------|------------|-------------|-----------|----------------------------|
| HTTP | TCP      | 80         | Anywhere    | 0.0.0.0/0 | e.g. SSH for admin desktop |

**Security group rule 2 (TCP, 22, 24.89.79.95/32)**

| Type | Protocol | Port range | Source type | Name           | Description - optional     |
|------|----------|------------|-------------|----------------|----------------------------|
| ssh  | TCP      | 22         | My IP       | 24.89.79.95/32 | e.g. SSH for admin desktop |

The right pane shows a **Summary** of the configuration:

- Software Image (AMI):** Amazon Linux 2023 AMI 2023.12.....[read more](#) (ami-Ofef201115eefe936)
- Virtual server type (instance type):** t2.micro
- Firewall (security group):** New security group
- Storage (volumes):** 1 volume(s) - 8 GiB

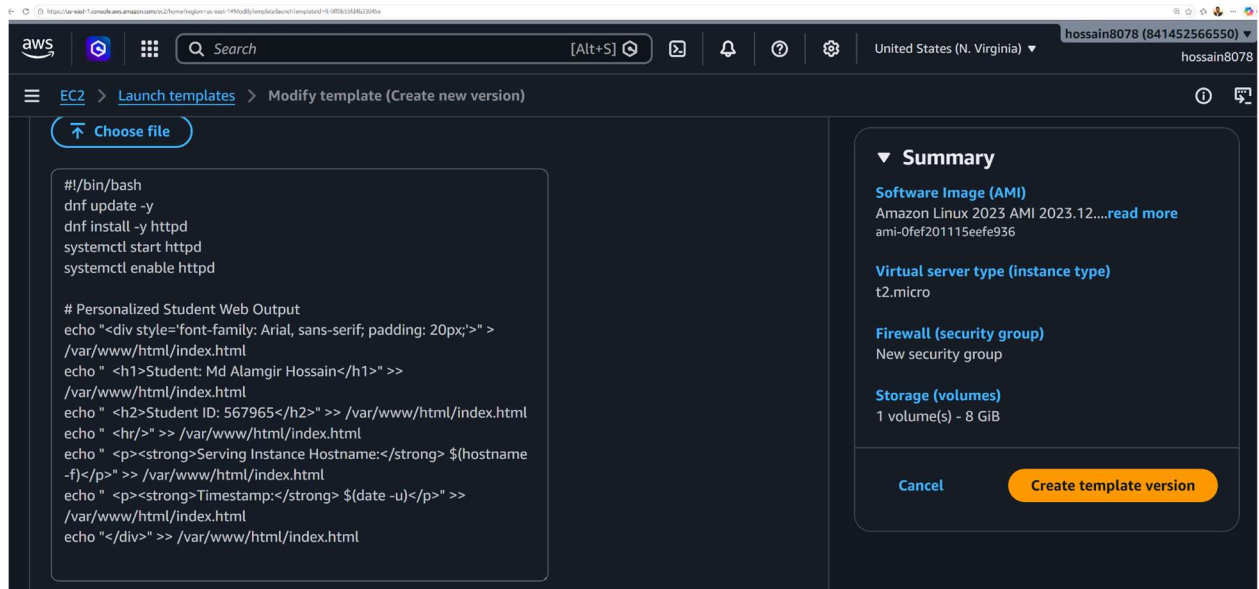
At the bottom right, there are two buttons: "Cancel" and "Create template version".

**Note:** Confirms HTTP port 80 is public and SSH port 22 is restricted to My IP.

## 6. Advanced Details (Personalized User Data Script):

- Expand the **Advanced details** section at the bottom of the page.
- Scroll down to **User data**.
- Copy and paste the script below, replacing the bracketed student placeholders with your exact details:

### Personalized User Data Script



```
#!/bin/bash
dnf update -y
dnf install -y httpd
systemctl start httpd
systemctl enable httpd

# Personalized Student Web Output
echo "<div style='font-family: Arial, sans-serif; padding: 20px;'>" > /var/www/html/index.html
echo "  <h1>Student: Md Alamgir Hossain</h1>" >> /var/www/html/index.html
echo "  <h2>Student ID: 567965</h2>" >> /var/www/html/index.html
echo "  <hr/>" >> /var/www/html/index.html
echo "  <p><strong>Serving Instance Hostname:</strong> ${hostname-f}</p>" >> /var/www/html/index.html
echo "  <p><strong>Timestamp:</strong> $(date -u)</p>" >> /var/www/html/index.html
echo "</div>" >> /var/www/html/index.html
```

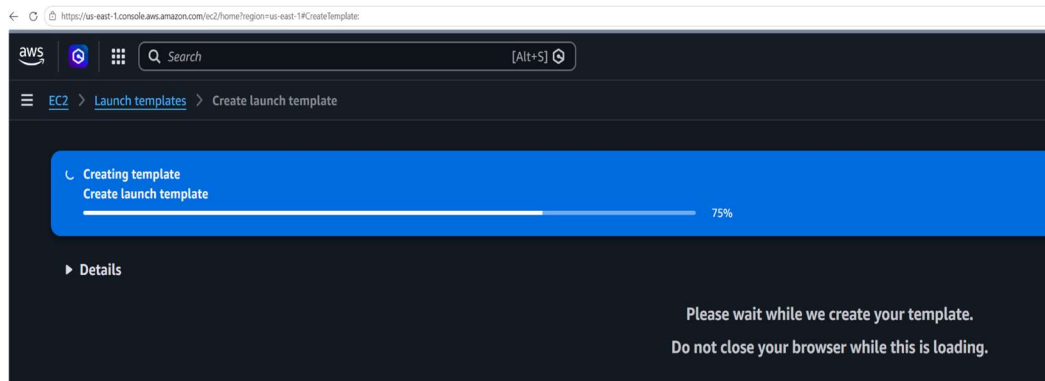
```
#!/bin/bash
dnf update -y
dnf install -y httpd
systemctl start httpd
systemctl enable httpd

# Personalized Student Web Output
echo "<div style='font-family: Arial, sans-serif; padding: 20px;'>" > /var/www/html/index.html
echo "  <h1>Student: [YOUR_FULL_NAME]</h1>" >> /var/www/html/index.html
echo "  <h2>Student ID: [YOUR_STUDENT_ID]</h2>" >> /var/www/html/index.html
echo "  <hr/>" >> /var/www/html/index.html
echo "  <p><strong>Serving Instance Hostname:</strong> $(hostname-f)</p>" >> /var/www/html/index.html
echo "  <p><strong>Timestamp:</strong> $(date -u)</p>" >> /var/www/html/index.html
echo "</div>" >> /var/www/html/index.html
```

**Note:** Confirms Apache installation and creation of the student identification web page are automated.

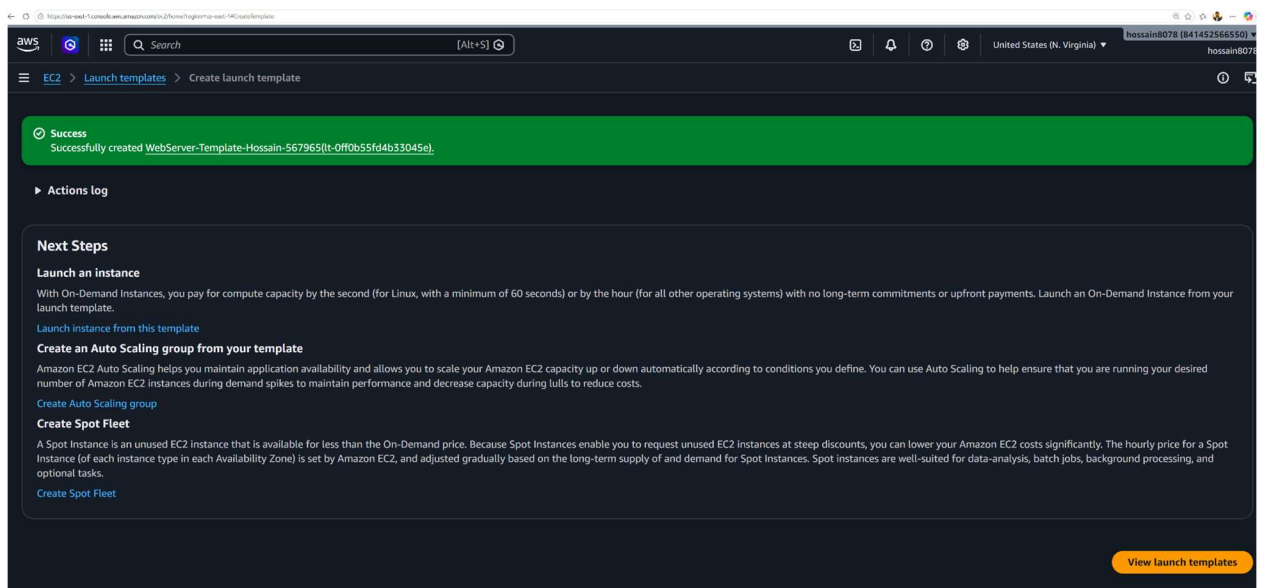
- Click **Create launch template**, then click **View Launch Templates**.

### Launch Template Creation in Progress



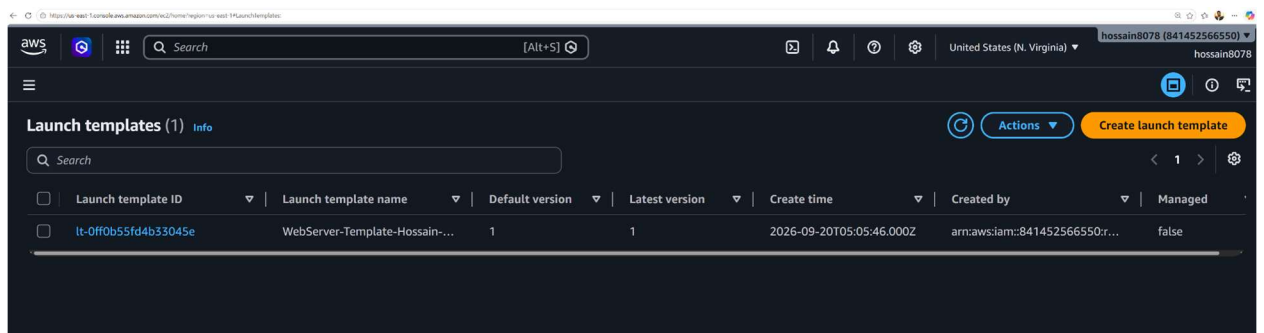
**Note:** Shows AWS processing the new launch template configuration.

### Launch Template Creation Confirmation



**Note:** Confirms the launch template was created successfully.

### Launch Template List



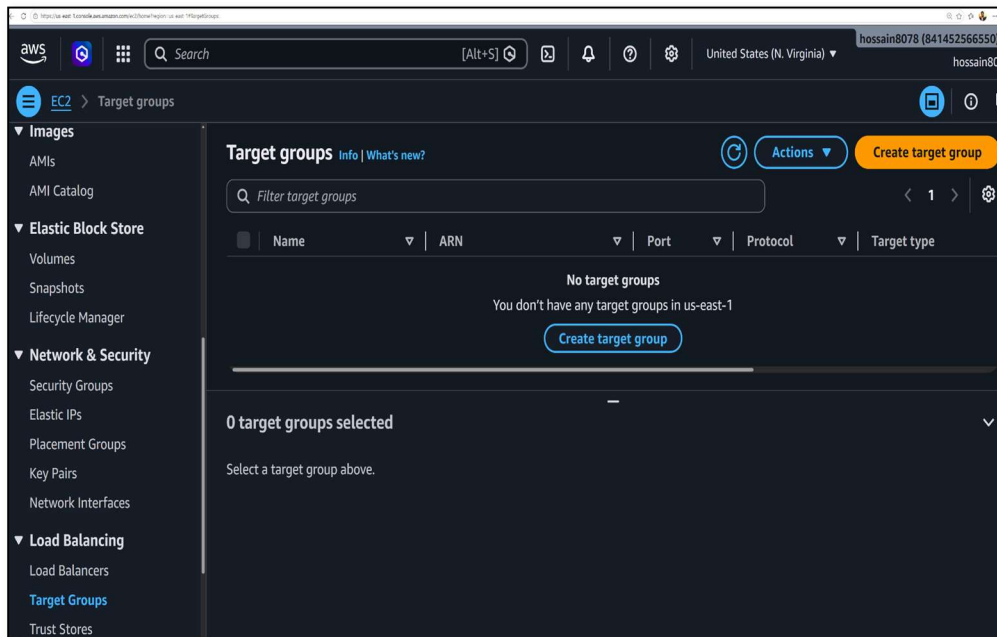
**Note:** Shows the personalized template available for use by an Auto Scaling group.

## Part 2: Configuring High Availability and Load Balancing

### Step 2.1: Create a Target Group

2.1.1 In the EC2 console left navigation, under **Load Balancing**, click **Target Groups**.

#### Target Groups Page

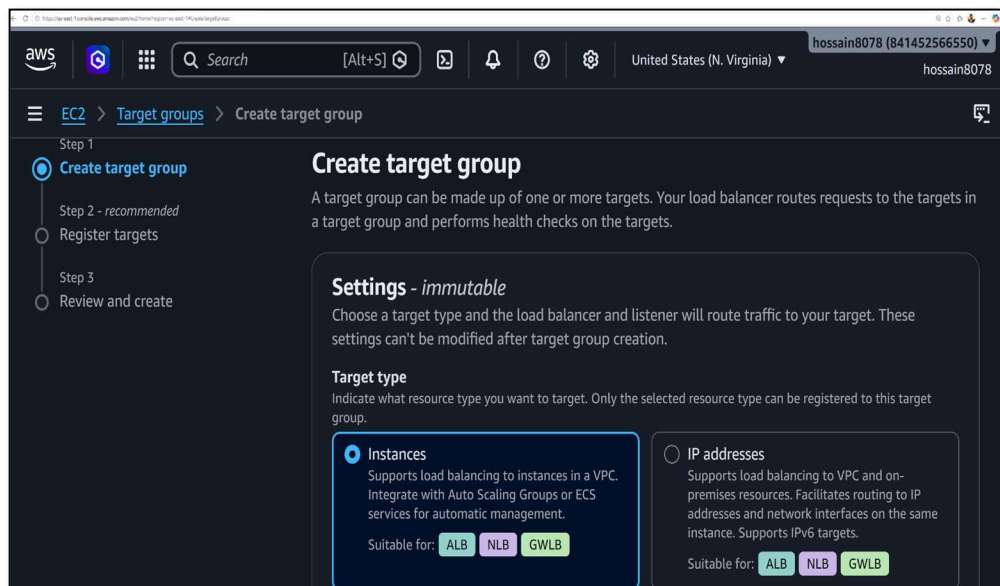


**Note:** Shows navigation to the Target Groups area under Load Balancing.

2.1.2 Click **Create target group**.

2.1.3 Under **Choose a target type**, select **Instances**.

#### Target Type Selection



**Note:** Confirms Instances is selected as the target type.

2.1.4 **Target group name:** WebServer-TG-[YourLastName]

2.1.5 Keep protocol as **HTTP** and port as **80**.

2.1.6 Ensure your **Default VPC** is selected. Leave all health check defaults intact.

### Target Group Network Settings

Search [Alt+S]

United States (N. Virginia) hossain8078 (841452566550) hossain8078

> Create target group

**Target group name**  
Name must be unique per Region per AWS account.  
WebServer-TG-Hossain  
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 20/32

**Protocol**  
Protocol for communication between the load balancer and targets.  
HTTP

**Port**  
Port number where targets receive traffic. Can be overridden for individual targets during registration.  
80  
1-65535

**IP address type**  
Only targets with the indicated IP address type can be registered to this target group.  
☒ IPv4  
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.  
☐ IPv6  
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

**VPC**  
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.  
vpc-0ab664410caf1682b (default)  
172.31.0.0/16  
[Create VPC](#)

**Note:** Confirms the target group name, HTTP port 80, IPv4, and default VPC.

2.1.7 Click **Next**.

2.1.8.1 On the **Register targets** page, **do not add targets manually**. The Auto Scaling Group will handle registration.

### Empty Target Registration

**Register targets - recommended**  
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

**Available instances (0)**  
Filter instances  
No instances

**0 selected**  
**Ports for the selected instances**  
Ports for routing traffic to the selected instances.  
80  
1-65535 (separate multiple ports with comma)  
[Include as pending below](#)

**Review targets**  
**Targets (0)**  
Filter targets  
Show only pending  
Remove all pending  
No instances added yet  
Specify instances above, or leave the group empty if you prefer to add targets later.  
0 pending  
[Cancel](#) [Previous](#) [Next](#)

**Note:** Confirms no instances were registered manually because Auto Scaling will register them later.

## 2.1.8.2 Click **Create target group**.

### Target Group Review

The screenshot shows the 'Review and create' step of the 'Create target group' wizard. The left sidebar indicates the current step is 'Review and create'. The main content area is titled 'Review and create' and includes a sub-header 'Step 1: Target group details'. Below this, there are two sections: 'Target group details' and 'Health check details'. The 'Target group details' section shows the Name 'WebServer-TG-Hossain', Target type 'Instance', Protocol : Port 'HTTP: 80', Protocol version 'HTTP1', VPC 'vpc-0ab664410caf1682b', and IP address type 'IPv4'. The 'Health check details' section shows Health check protocol 'HTTP', Health check path '/', Health check port 'traffic-port', Interval '30 seconds', Timeout '5 seconds', Healthy threshold '5', Unhealthy threshold '2', and Success codes '200'. Below these sections is 'Step 2: Register targets', which shows a table with 0 targets. At the bottom, there are buttons for 'Cancel', 'Previous', and 'Create target group'.

Step 1  
● Create target group  
● Step 2 - recommended  
● Register targets  
● Step 3  
● **Review and create**

### Review and create

Review your target group configuration before creating

#### Step 1: Target group details

**Target group details**

|                                              |                         |                             |                           |
|----------------------------------------------|-------------------------|-----------------------------|---------------------------|
| Name<br>WebServer-TG-Hossain                 | Target type<br>Instance | Protocol : Port<br>HTTP: 80 | Protocol version<br>HTTP1 |
| VPC<br><a href="#">vpc-0ab664410caf1682b</a> | IP address type<br>IPv4 |                             |                           |

**Health check details**

|                               |                        |                                   |                        |
|-------------------------------|------------------------|-----------------------------------|------------------------|
| Health check protocol<br>HTTP | Health check path<br>/ | Health check port<br>traffic-port | Interval<br>30 seconds |
| Timeout<br>5 seconds          | Healthy threshold<br>5 | Unhealthy threshold<br>2          | Success codes<br>200   |

#### Step 2: Register targets

**Targets (0)**

| Instance ID      | Name | Port | Zone |
|------------------|------|------|------|
| No targets added |      |      |      |

**Server-side tasks and status**  
After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

Cancel Previous **Create target group**

**Note:** Confirms the HTTP health check and empty target list before creation.

### Created Target Group

The screenshot shows the 'WebServer-TG-Hossain' target group details page in the AWS Management Console. The left sidebar shows the navigation menu with 'Network & Security' and 'Load Balancing' expanded. The main content area shows the 'Details' tab for the target group. The details include the ARN, Target type 'Instance', Protocol : Port 'HTTP: 80', Protocol version 'HTTP1', VPC 'vpc-0ab664410caf1682b', and IP address type 'IPv4'. Below the details, there is a summary row showing 0 Total targets, 0 Healthy, 0 Unhealthy, 0 Unused, 0 Initial, and 0 Draining. At the bottom, there are tabs for 'Targets', 'Monitoring', 'Health checks', 'Attributes', and 'Tags'.

aws Search [Alt+S] United States (N. Virginia) hossain8078

EC2 > Target groups > WebServer-TG-Hossain

### WebServer-TG-Hossain

**Details**

arn:aws:elasticloadbalancing:us-east-1:841452566550:targetgroup/WebServer-TG-Hossain/62cbf8a8b8f90acb

|                         |                                                  |                           |                                              |
|-------------------------|--------------------------------------------------|---------------------------|----------------------------------------------|
| Target type<br>Instance | Protocol : Port<br>HTTP: 80                      | Protocol version<br>HTTP1 | VPC<br><a href="#">vpc-0ab664410caf1682b</a> |
| IP address type<br>IPv4 | Load balancer<br><a href="#">None associated</a> |                           |                                              |

0 Total targets    0 Healthy    0 Unhealthy    0 Unused    0 Initial    0 Draining

0 Anomalous

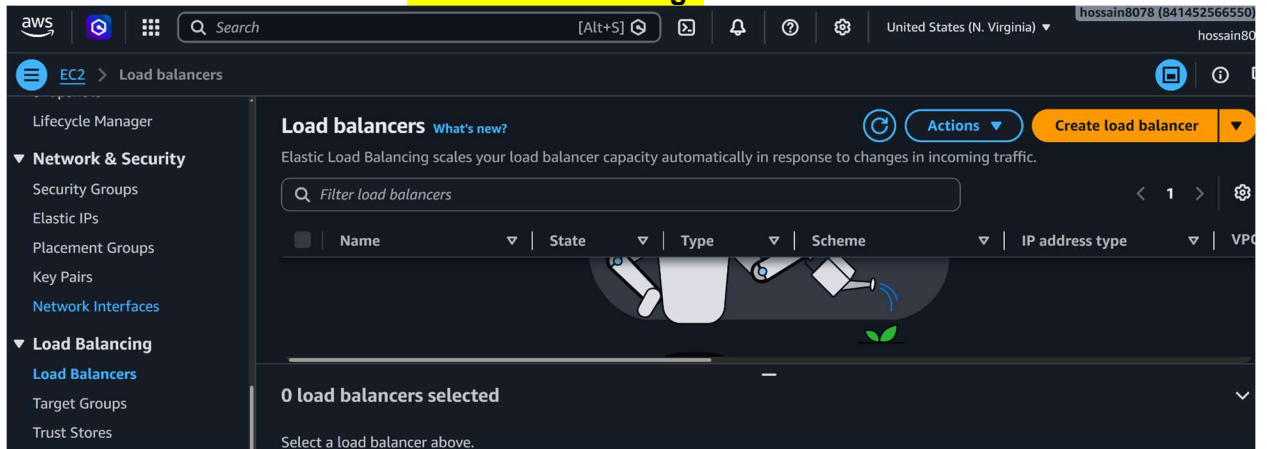
**Targets** Monitoring Health checks Attributes Tags

**Note:** Confirms the target group exists and is ready for Auto Scaling targets.

## Step 2.2: Deploy the Application Load Balancer (ALB)

2.2.1 In the left navigation, click **Load Balancers** > **Create load balancer**.

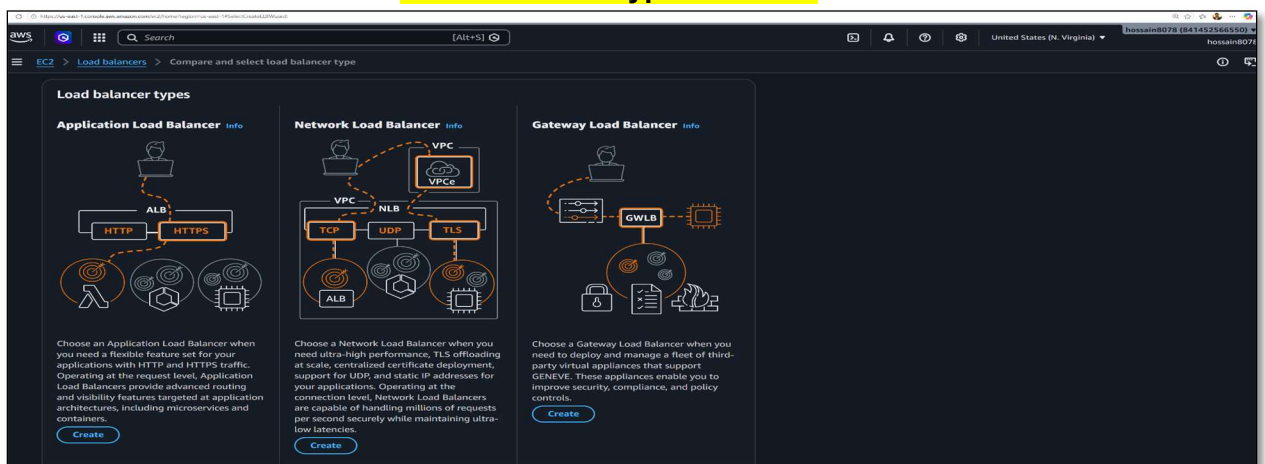
### Load Balancers Page



**Note:** Shows the starting point for creating the Application Load Balancer.

2.2.2 Under **Application Load Balancer**, click **Create**.

### Load Balancer Type Selection



**Note:** Confirms that Application Load Balancer is selected.

### 2.2.3 Basic Configuration:

- **Name:** Web-ALB-[YourLastName]
- **Scheme:** Internet-facing
- **IP address type:** IPv4

#### ALB Basic Configuration

The screenshot shows the 'Basic configuration' tab of the 'Create Application Load Balancer' page in the AWS Management Console. The 'Load balancer name' field is set to 'Web-ALB-Hossain'. The 'Scheme' is set to 'Internet-facing'. The 'Load balancer IP address type' is set to 'IPv4'. The page includes instructions for naming conventions and details about the selected scheme and IP address type.

**Basic configuration**

**Load balancer name**  
Name must be unique within your AWS account and can't be changed after the load balancer is created.  
Web-ALB-Hossain  
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

**Scheme** | Info  
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves Internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

**Load balancer IP address type** | Info  
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**  
Includes only IPv4 addresses.

☐ **Dualstack**  
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

**Note:** Confirms the personalized name, internet-facing scheme, and IPv4 address type.

### 2.2.4 Network Mapping:

- **VPC:** Select your **Default VPC**.
- **Mappings:** Select at least **two** Availability Zones (e.g., us-east-1a and us-east-1b).  
For each selected zone, pick the default public subnet presented in the dropdown.

#### ALB Network Mapping

The screenshot shows the 'Network mapping' tab of the 'Create Application Load Balancer' page. The 'VPC' is set to 'vpc-0ab664410caf1682b'. The 'IP pools' section has 'Use IPAM pool for public IPv4 addresses' checked. Two Availability Zones are selected: 'us-east-1a (use1-az1)' and 'us-east-1b (use1-az2)', each with a corresponding public subnet selected from a dropdown menu.

**Network mapping** | Info  
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

**VPC** | Info  
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0ab664410caf1682b (default) Create VPC

**IP pools** | Info  
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☒ **Use IPAM pool for public IPv4 addresses**  
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

**Availability Zones and subnets** | Info  
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az1)**

Subnet  
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0955a41b19a48ef4a  
172.31.0.0/20  
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ **us-east-1b (use1-az2)**

Subnet  
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

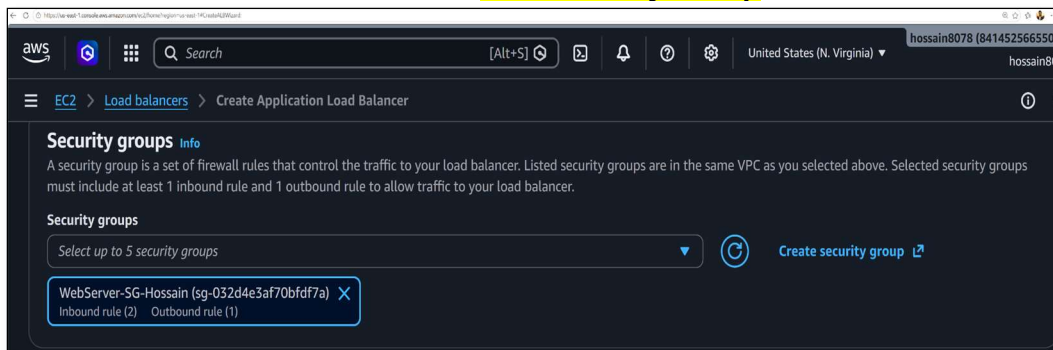
subnet-099aa8df5dea7ceb5  
172.31.80.0/20  
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

**Note:** Confirms the default VPC and public subnets in two Availability Zones.

### 2.2.5 Security Groups:

- Remove the default security group.
- Select WebServer-SG-[YourLastName].

#### ALB Security Group

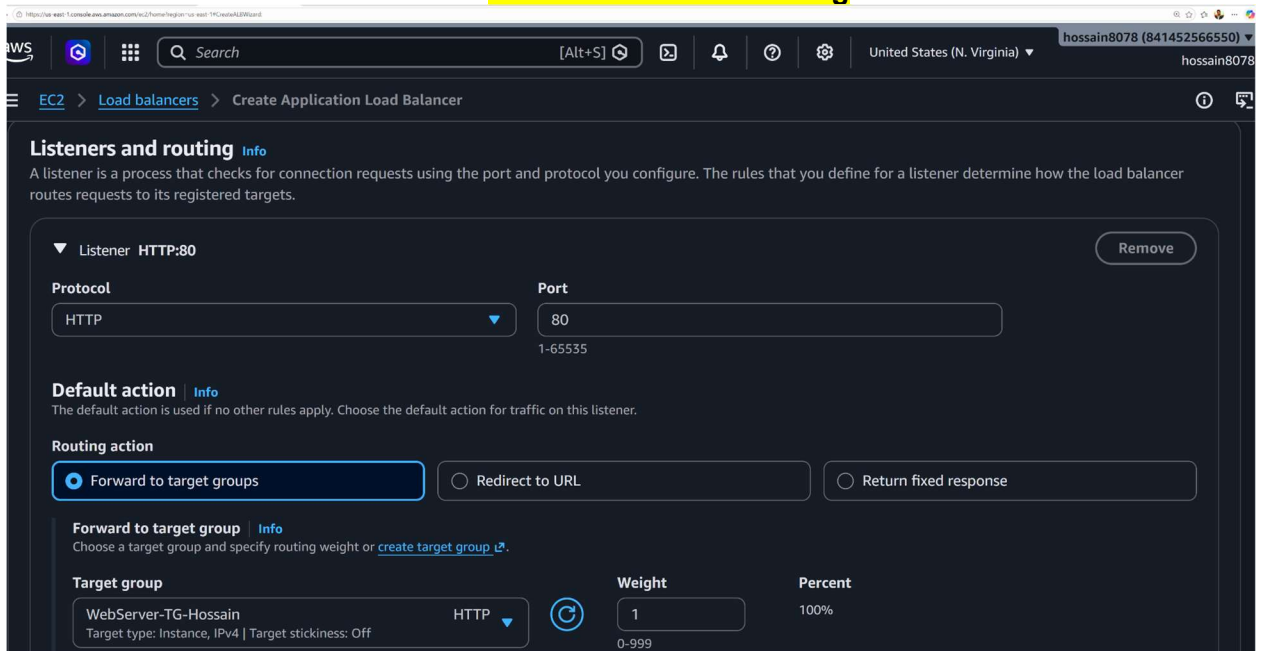


**Note:** Confirms the web server security group is assigned to the load balancer.

### 2.2.6 Listeners and Routing:

- Protocol: **HTTP** | Port: **80**
- Default action: Select **Forward to** and choose WebServer-TG-[YourLastName].

#### ALB Listener and Routing



**Note:** Confirms the HTTP port 80 listener forwards requests to the web server target group.

## 2.2.7 Click **Create load balancer**.

### ALB Review and Create

The screenshot shows the 'Review' step of the 'Create Application Load Balancer' wizard. The page is divided into several sections for configuration review:

- Summary:** Review and confirm your configurations. [Estimate cost](#)
- Basic configuration:** Name: Web-ALB-Hossain, Scheme: Internet-facing, IP address type: IPv4.
- Network mapping:** VPC: vpc-0ab664410caf1682b, Public IPv4 IPAM pool: -, Availability Zones and subnets: us-east-1a (subnet-0955a41b19a48ef4a), us-east-1b (subnet-099aa8df5dea7ceb5).
- Security groups:** WebServer-SG-Hossain, sg-032d4e3af70bfd7a.
- Listeners and routing:** HTTP:80 | Forward to 1 target group.
- Service integrations:** Amazon CloudFront + AWS Web Application Firewall (WAF): -, AWS WAF: -, AWS Global Accelerator: -.
- Attributes:** A message states: 'Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.'
- Creation workflow and status:** A section for 'Server-side tasks and status' with a message: 'After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.'

At the bottom right, there are two buttons: 'Cancel' and 'Create load balancer'.

**Note:** Summarizes the load balancer settings before creation.

### ALB Creation Confirmation

The screenshot shows the 'Web-ALB-Hossain' page in the AWS Management Console. At the top, there are two status messages:

- Successfully created load balancer: Web-ALB-Hossain** (green bar): It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.
- Introducing new Capacity Utilization metric for ALB** (blue bar): Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the [Capacity](#) or [Monitoring](#) tabs to view utilization data.

The main section is titled 'Web-ALB-Hossain' and includes an 'Actions' button. Below this is a 'Details' section with the following information:

- Load balancer type:** Application
- Status:** Provisioning
- VPC:** vpc-0ab664410caf1682b
- Load balancer IP address type:** IPv4
- Scheme:** Internet-facing
- Hosted zone:** Z355XDOTRQ7X7K
- Availability Zones:** subnet-0955a41b19a48ef4a (us-east-1a (use1-az1)), subnet-099aa8df5dea7ceb5 (us-east-1b (use1-az2))
- Date created:** September 19, 2026, 23:58 (UTC-06:00)
- Load balancer ARN:** arn:aws:elasticloadbalancing:us-east-1:841452566550:loadbalancer/app/Web-ALB-Hossain/92d87615065a3b72
- DNS name info:** Web-ALB-Hossain-1917833292-us-east-1.elb.amazonaws.com (A Record)

Below the details is a tabbed interface with 'Listeners and rules' selected. It shows a table of listeners and rules:

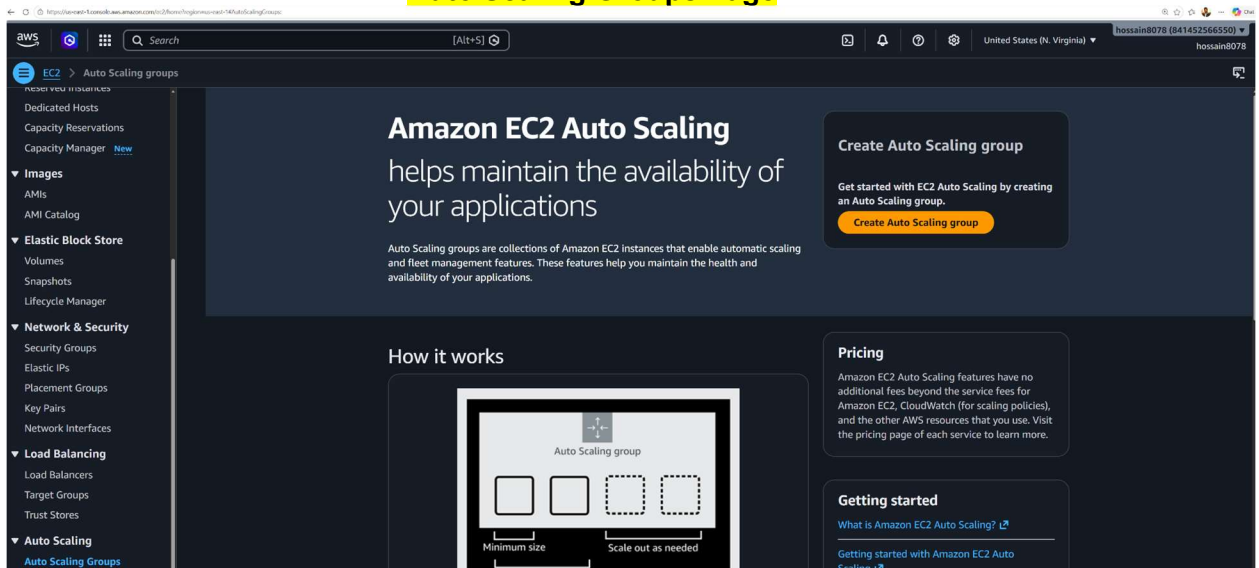
| Protocol/Port | Default action                                                                      | Rules  | ARN | Security policy | Default SSL/TLS certificate | mTLS           | Trust store |
|---------------|-------------------------------------------------------------------------------------|--------|-----|-----------------|-----------------------------|----------------|-------------|
| HTTP:80       | Forward to target group WebServer-TG-Hossain (100%)<br>Target group stickiness: Off | 1 rule | ARN | Not applicable  | Not applicable              | Not applicable | Not applic  |

**Note:** Confirms the load balancer was created and initially entered the Provisioning state.

## Step 2.3: Configure the Auto Scaling Group (ASG)

In the left navigation, under **Auto Scaling**, click **Auto Scaling Groups** > **Create Auto Scaling group**.

### Auto Scaling Groups Page

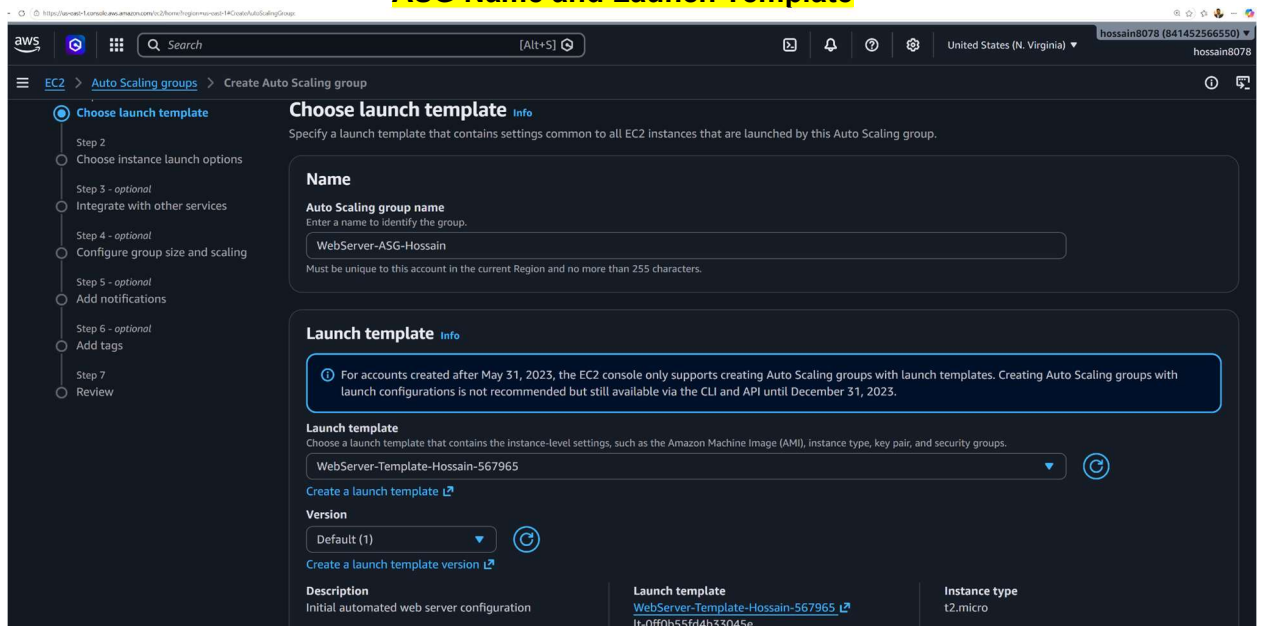


**Note:** Shows the starting point for creating an Auto Scaling group.

## Step 1 - Choose launch template:

- **Auto Scaling group name:** WebServer-ASG-[YourLastName]
- **Launch template:** Select WebServer-Template-[YourLastName]-[StudentID].

### ASG Name and Launch Template



**Note:** Confirms the personalized Auto Scaling group name and launch template selection.

- Click **Next**.

## Launch Template Details

The screenshot shows the 'Choose launch template' step in the AWS Management Console. The left sidebar lists steps from 1 to 7, with 'Step 1: Choose launch template' selected. The main content area is titled 'Choose launch template' and includes a description: 'Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.' Below this, there are three main sections: 'Name', 'Launch template', and 'Additional details'. The 'Name' section has a text input field for 'Auto Scaling group name' with the value 'WebServer-ASG-Hossain'. The 'Launch template' section has a dropdown menu for 'Launch template' with the value 'WebServer-Template-Hossain-567965'. The 'Additional details' section shows 'AMI ID' as 'ami-0fe201115eef936', 'Key pair name' as 'CCMP200T05', and 'Instance type' as 't2.micro'. At the bottom right, there are 'Cancel' and 'Next' buttons.

**Note:** Confirms the default template version and t2.micro instance type.

## Step 2 - Network:

- **VPC:** Ensure your **Default VPC** is selected.
- **Subnets:** Select the subnets that match the two Availability Zones chosen for your ALB. Click **Next**.

## ASG VPC and Subnets

The screenshot shows the 'Choose instance launch options' step in the AWS Management Console. The left sidebar lists steps from 1 to 7, with 'Step 2: Choose instance launch options' selected. The main content area is titled 'Choose instance launch options' and includes a description: 'Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.' Below this, there are three main sections: 'Instance type requirements', 'Network', and 'Availability Zones and subnets'. The 'Instance type requirements' section has a dropdown menu for 'Launch template' with the value 'WebServer-Template-Hossain-567965' and a dropdown menu for 'Instance type' with the value 't2.micro'. The 'Network' section has a dropdown menu for 'VPC' with the value 'vpc-dab66441caf1682b'. The 'Availability Zones and subnets' section has a dropdown menu for 'Select Availability Zones and subnets' with the value 'use1-az1 (us-east-1a) | subnet-0955a41b19a48ef4a'. At the bottom right, there are 'Cancel' and 'Next' buttons.

**Note:** Confirms the Auto Scaling group uses two subnets in different Availability Zones.

## Instance Launch Options

The screenshot shows the 'Choose instance launch options' step in the AWS Management Console. The left sidebar lists steps: Step 1: Choose launch template, Step 2: Choose instance launch options (selected), Step 3: optional: Integrate with other services, Step 4: optional: Configure group size and scaling, Step 5: optional: Add notifications, Step 6: optional: Add tags, and Step 7: Review. The main content area is titled 'Choose instance launch options' and includes sections for 'Instance type requirements', 'Launch template' (Amazon Linux 2 AMI), 'Instance type' (t3.micro), 'Network' (VPC: vpc-0a6654410af16c3b, Subnets: subnet-093d4110175adb1fa, subnet-093d4110175adb1fa), 'Availability Zones and subnets' (us-east-1a, us-east-1b), 'Availability Zone distribution' (Balanced best effort selected), and 'Additional capacity settings' (Capacity Reservation preference: Default selected).

**Note:** Summarizes the selected VPC, subnets, and default capacity settings.

### Step 3 - Advanced Options:

- Under **Load balancing**, select **Attach to an existing load balancer**.
- Select **Choose from your load balancer target groups** and select WebServer-TG-[YourLastName].

## Target Group Attachment

The screenshot shows the 'Integrate with other services - optional' step in the AWS Management Console. The left sidebar lists steps: Step 1: Choose launch template, Step 2: Choose instance launch options, Step 3: optional: Integrate with other services (selected), Step 4: optional: Configure group size and scaling, Step 5: optional: Add notifications, Step 6: optional: Add tags, and Step 7: Review. The main content area is titled 'Integrate with other services - optional' and includes sections for 'Load balancing' (Attach to an existing load balancer selected), 'Attach to an existing load balancer' (Choose from your load balancer target groups selected), and 'Existing load balancer target groups' (WebServer-TG-Hossain | HTTP selected).

**Note:** Confirms the Auto Scaling group is attached to the existing web server target group.

- Under **Health checks**, check the box for **Turn on Elastic Load Balancing health checks**. Click **Next**.

## Load Balancing Health Checks

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

**Select VPC Lattice service to attach**

☒ **No VPC Lattice service**  
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ **Attach to VPC Lattice service**  
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

**Application Recovery Controller (ARC) zonal shift - new** [info](#)  
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ **Enable zonal shift**  
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

**Health checks**  
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

**EC2 health checks**  
[Always enabled](#)

**Additional health check types - optional** [info](#)  
**Turn on Elastic Load Balancing health checks** [Recommended](#)  
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☒ **EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing.** To avoid unexpected terminations, first verify the settings of these health checks in the [Load Balancer console](#).

☐ **Turn on VPC Lattice health checks**  
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ **Turn on Amazon EBS health checks**  
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

**Health check grace period** [info](#)  
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

**Note:** Confirms Elastic Load Balancing health checks are enabled with a grace period.

## Step 4 - Group Size:

- **Desired capacity: 2**
- **Minimum capacity: 2**
- **Maximum capacity: 4**

## ASG Capacity Settings

**Configure group size and scaling - optional** [info](#)  
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

**Group size** [info](#)  
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

**Desired capacity type**  
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

**Desired capacity**  
Specify your group size.

2

**Scaling** [info](#)  
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

**Scaling limits**  
Set limits on how much your desired capacity can be increased or decreased.

**Min desired capacity**  
2

**Max desired capacity**  
4

Equal or less than desired capacity      Equal or greater than desired capacity

**Automatic scaling - optional**  
**Choose whether to use a target tracking policy** [info](#)  
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ **No scaling policies**  
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ **Target tracking scaling policy**  
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

**Note:** Confirms desired capacity 2, minimum 2, maximum 4, and no scaling policy.

- Click **Next**.

Click **Next** through *Add notifications* and *Add tags* without changes.

## Notifications Configuration

The screenshot shows the 'Add notifications' step in the AWS Management Console. On the left, a progress bar indicates steps 1 through 5, with 'Add notifications' selected. The main area is titled 'Add notifications - optional' and includes a description: 'Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.' Below this is an 'Add notification' button. At the bottom right, there are 'Cancel', 'Skip to review', 'Previous', and 'Next' buttons. The 'Next' button is highlighted in orange.

**Note:** Confirms no optional notification was added.

## ASG Tags Configuration

The screenshot shows the 'Add tags' step in the AWS Management Console. The progress bar on the left shows steps 1 through 7, with 'Add tags' selected. The main area is titled 'Add tags - optional' and includes a description: 'Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.' Below this is a warning box stating: 'You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.' Below the warning is a section titled 'Tags (0)' with 'No tags' and an 'Add tag' button. At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons. The 'Next' button is highlighted in orange.

**Note:** Confirms no additional Auto Scaling group tag was added at this step.

## ASG Review

The screenshot shows the 'Review' step in the AWS Management Console. The progress bar on the left shows steps 1 through 7, with 'Review' selected. The main area is titled 'Review' and includes a description: 'Review the configuration for your Auto Scaling group before creating it.' Below this are two sections: 'Step 1: Choose launch template' and 'Step 2: Choose instance launch options'. The 'Step 1' section shows 'Group details' with 'Auto Scaling group name' as 'WebServer-ASG-Hossain' and 'Launch template' as 'WebServer-Template-Hossain-567965'. The 'Step 2' section shows 'Network' with 'VPC' as 'vpc-0ab664410caf1682b' and 'Availability Zones and subnets' as 'us-east-1a' and 'us-east-1b'. At the bottom right, there are 'Edit' buttons for both steps. The 'Next' button is highlighted in orange.

| Availability Zone | Subnet                   | Subnet CIDR range |
|-------------------|--------------------------|-------------------|
| us-east-1a        | subnet-0955a41b19a48ef4a | 172.31.0.0/20     |
| us-east-1b        | subnet-099aa8df5dea7ceb5 | 172.31.80.0/20    |

**Note:** Shows the complete Auto Scaling configuration before creation.

Review settings and click **Create Auto Scaling group**.

### Create Auto Scaling Group

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console, specifically Step 6: Add tags. The page has a dark theme. At the top, there's a navigation bar with the AWS logo, a search bar, and the user's account information (hossain8078). The breadcrumb trail shows 'EC2 > Auto Scaling groups > Create Auto Scaling group'. Below the breadcrumb, there's a 'Notifications' section with 'No notifications'. The main section is 'Step 6: Add tags' with an 'Edit' button. It contains a 'Tags (0)' section with a table for adding tags. The table has columns for 'Key', 'Value', and 'Tag new instances'. Below the table, it says 'No tags'. At the bottom, there are buttons for 'Preview code', 'Cancel', 'Previous', and 'Create Auto Scaling group'.

| Key     | Value | Tag new instances |
|---------|-------|-------------------|
| No tags |       |                   |

**Note:** Confirms the final configuration is ready to create the Auto Scaling group.

### ASG Updating Capacity

The screenshot shows the 'Auto Scaling groups' page in the AWS Management Console. The page has a dark theme. At the top, there's a navigation bar with the AWS logo, a search bar, and the user's account information (hossain8078). The breadcrumb trail shows 'EC2 > Auto Scaling groups'. Below the breadcrumb, there's a section for 'Auto Scaling groups (1)' with an 'Info' link. It shows 'Last updated less than a minute ago' and buttons for 'Launch configurations', 'Launch templates', 'Actions', and 'Create Auto Scaling group'. Below this, there's a search bar for 'Search your Auto Scaling groups'. At the bottom, there's a table with columns for 'Name', 'Status', 'Launch template/configuration', and 'Instances'. The table contains one entry: 'WebServer-ASG-Hossain' with status 'Updating capacity', launch template 'WebServer-Template-Hossain-567965', and 0 instances.

| Name                  | Status            | Launch template/configuration     | Instances |
|-----------------------|-------------------|-----------------------------------|-----------|
| WebServer-ASG-Hossain | Updating capacity | WebServer-Template-Hossain-567965 | 0         |

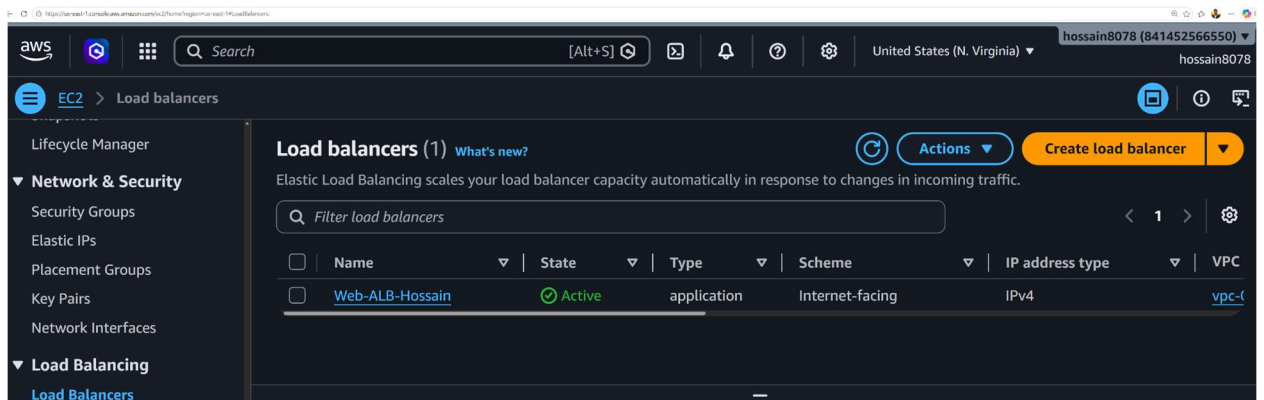
**Note:** Shows the new Auto Scaling group beginning to launch its desired instances.

## Part 3: Testing, Verification, and Fault Tolerance

### Step 3.1: Verify ALB Traffic Routing

-Go to **EC2 > Load Balancers** and select Web-ALB-[YourLastName].

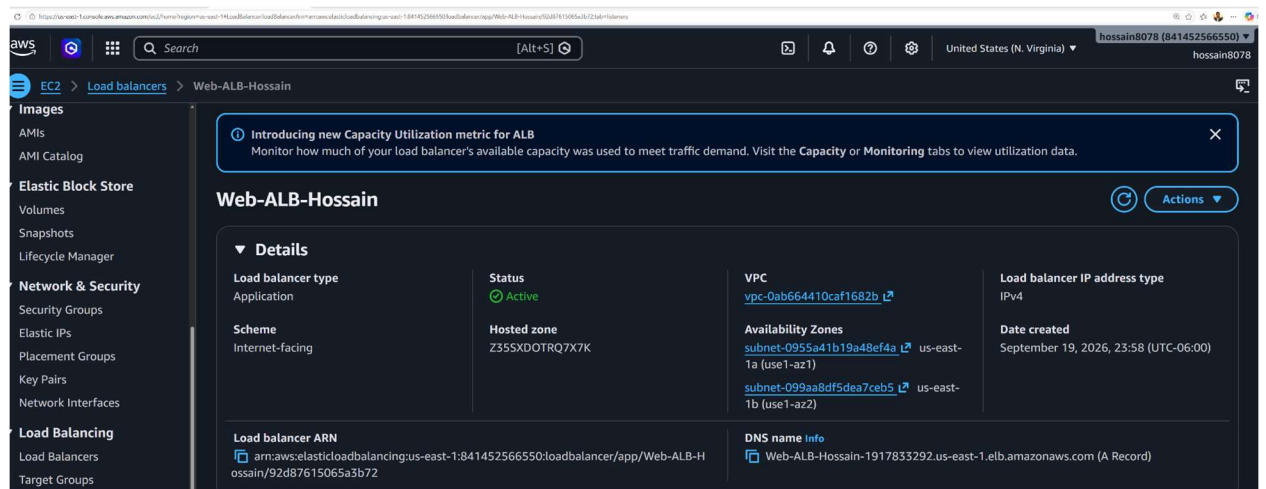
#### Active Load Balancer



**Note:** Confirms the Application Load Balancer reached the Active state.

-Ensure the **State** is **Active** (not *Provisioning*).

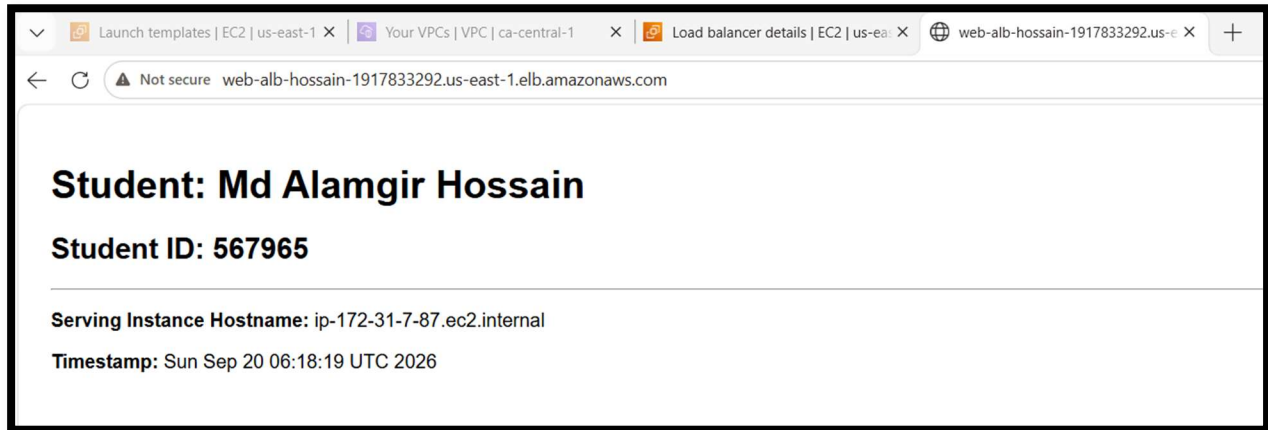
#### ALB Details and DNS Name



**Note:** Confirms the active internet-facing load balancer, two Availability Zones, and public DNS name.

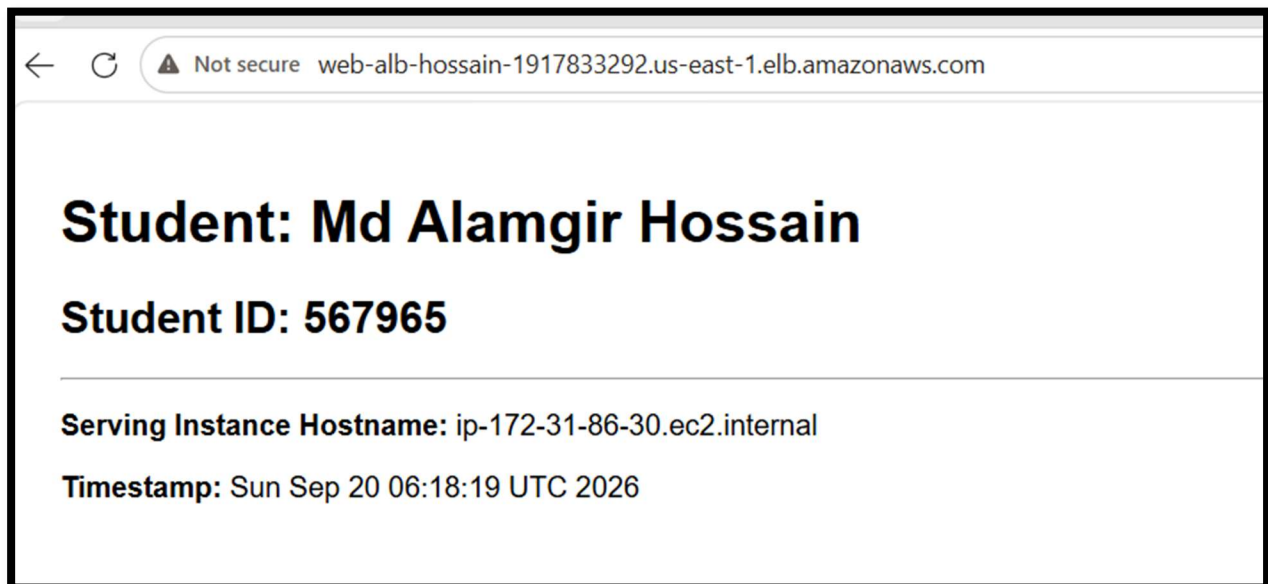
- Copy the **DNS name** (e.g., web-alb-smith-123456789.us-east-1.elb.amazonaws.com).
- Open a web browser and paste the URL.

### ALB Routing Result Instance A



**Note:** Shows the personalized page served through the ALB by instance hostname ip-172-31-7-87.

### ALB Routing Result Instance B

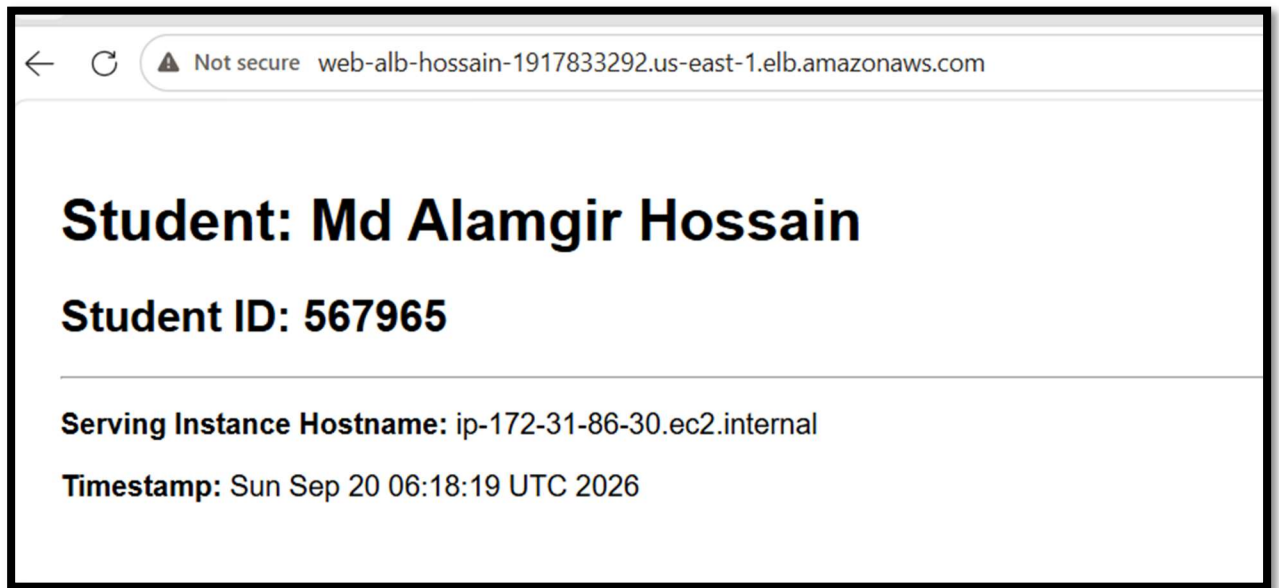


**Note:** Shows the same page served through the ALB by instance hostname ip-172-31-86-30.

**CRITICAL NOTE (HTTP vs. HTTPS):** Modern web browsers will automatically prepend https:// if you only enter the bare domain name. Because this lab deploys an unencrypted HTTP listener on port 80, loading https://... will cause an immediate connection drop or ERR\_TIMED\_OUT. You **must** explicitly prepend **http://** in your browser's address bar: http://Web-ALB-[YourLastName]-1234567890.us-east-1.elb.amazonaws.com

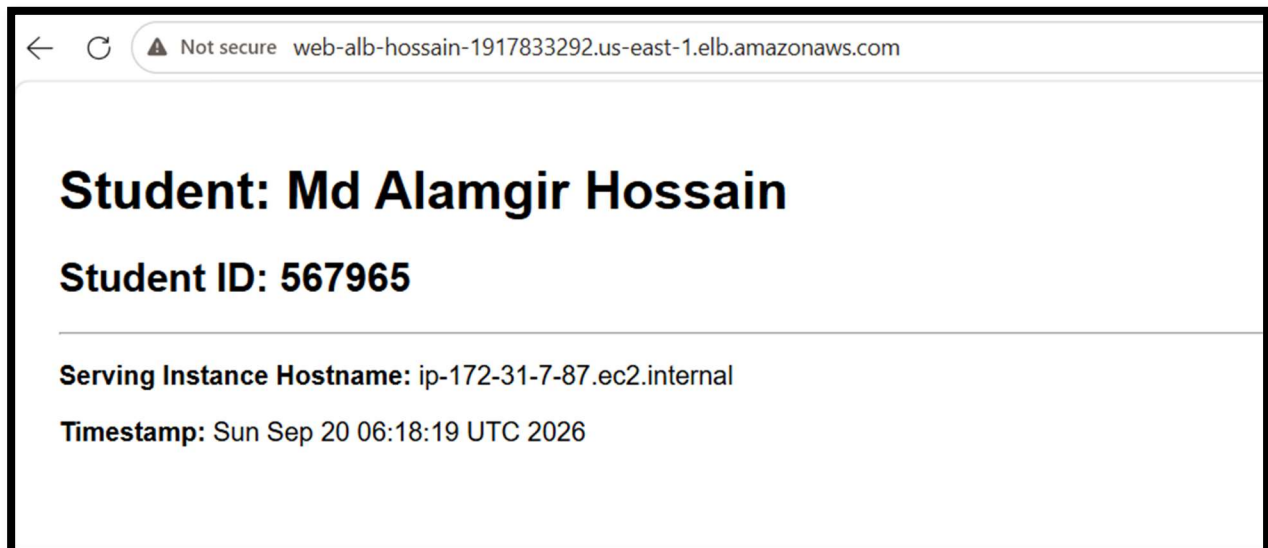
1. Verify that your personalized web page loads showing your **Name**, **Student ID**, and the serving instance's hostname.
2. Refresh the browser page repeatedly (or run `curl http://<ALB-DNS-NAME>` from a terminal). Observe that the **Hostname** flips between the two EC2 instances running behind the load balancer.

### Repeated Response from Instance B



**Note:** A refreshed request reached instance ip-172-31-86-30, demonstrating load-balanced routing.

### Repeated Response from Instance A

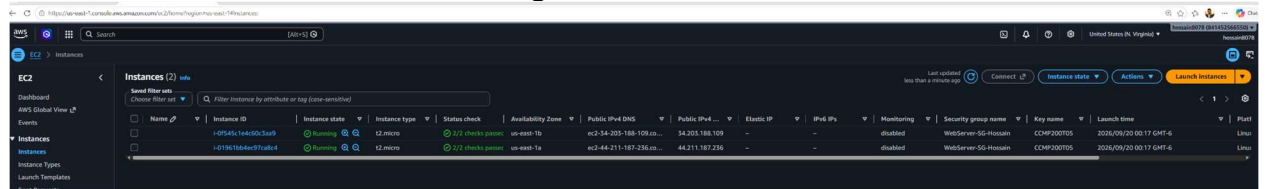


**Note:** A later request reached instance ip-172-31-7-87 while preserving identical student information.

## Step 3.2: Simulate Infrastructure Failure

1. Navigate to **EC2 > Instances**.

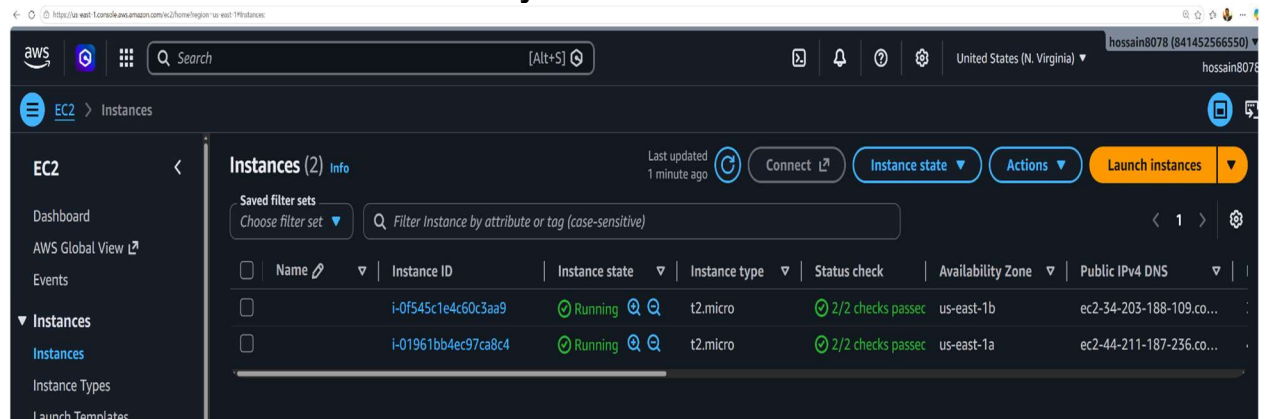
### ASG Managed Instances



**Note:** Shows the EC2 instances launched and managed by the Auto Scaling group.

2. Locate the two instances managed by your ASG.

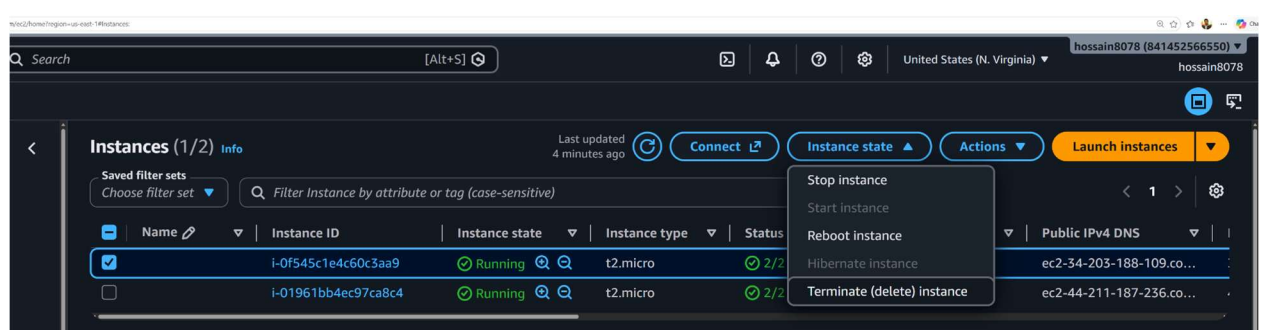
### Two Healthy EC2 Instances



**Note:** Confirms both instances are running with 2/2 status checks across two Availability Zones.

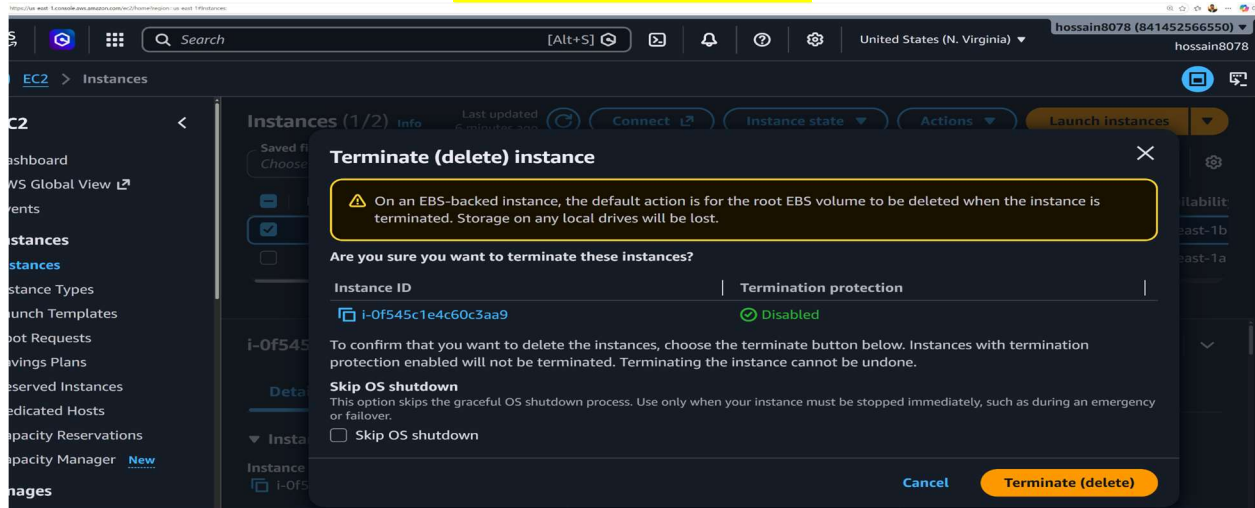
3. Select **one** of the instances, click **Instance state > Terminate instance**, and confirm.

### Terminate Instance Action



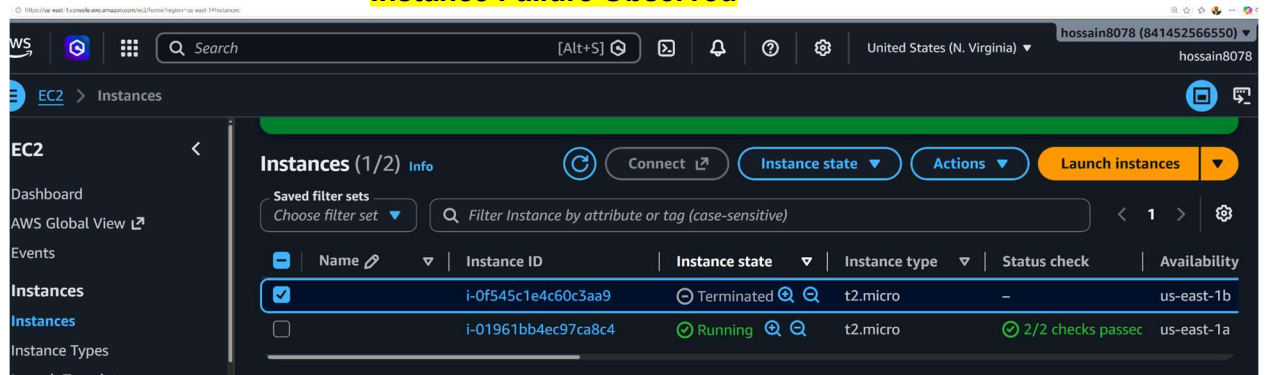
**Note:** Shows one managed instance selected for the failure simulation.

## Termination Confirmation



**Note:** Confirms the selected instance is about to be terminated deliberately.

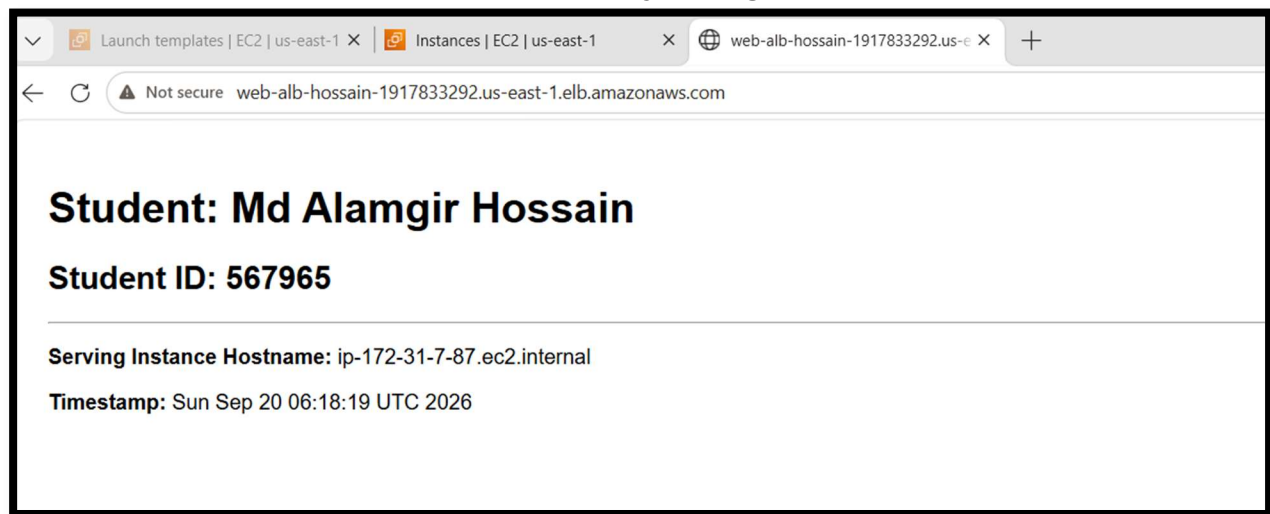
## Instance Failure Observed



**Note:** Shows one instance terminated while the second instance remains healthy.

4. Refresh your browser pointing at your ALB DNS name. Your website should stay online without downtime because the ALB automatically detects the failing target and routes all traffic to the remaining healthy instance.

## Website Availability During Failure



**Note:** Confirms the ALB-hosted page remains available through the surviving instance.

5. In the EC2 console, observe the recovery process:
- The terminated instance moves to Terminated.
  - Within 2 to 3 minutes, the ASG detects that active capacity is below 2 and launches a fresh instance.

**Launches a fresh instance**

The screenshot shows the AWS Management Console for the EC2 service. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, and Capacity Manager. The main content area displays the 'Instances (3)' page. At the top, there are buttons for 'Connect', 'Instance state', 'Actions', and 'Launch instances'. Below these is a search bar and a table of instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, and Availability. Three instances are listed: i-0350d70f5eee1d0b4 (Running), i-0f545c1e4c60c3aa9 (Terminated), and i-01961bb4ec97ca8c4 (Running). Below the table, the 'All instances' section shows three panels: 'Instance alarm states' (None), 'Service health' (Operating normally), and 'Health events affecting EC2 instances' (None).

| Name | Instance ID         | Instance state | Instance type | Status check      | Availability |
|------|---------------------|----------------|---------------|-------------------|--------------|
|      | i-0350d70f5eee1d0b4 | Running        | t2.micro      | 2/2 checks passed | us-east-1b   |
|      | i-0f545c1e4c60c3aa9 | Terminated     | t2.micro      | -                 | us-east-1b   |
|      | i-01961bb4ec97ca8c4 | Running        | t2.micro      | 2/2 checks passed | us-east-1a   |

**All instances**

- Instance alarm states**  
In alarm: None
- Service health**  
EC2 service status: Operating normally
- Health events affecting EC2 instances**  
Open events: None  
Upcoming events: None